Proposed CSA: category-level word–region supervision

Frozen CLIP ROI cache
R: B × N × 768 + valid mask

TRAIN-only annotations
Captions + GT category boxes
Cached proposal boxes / row IDs

Existing projection 768 → 512
CSA object1 → object2 → relation
Frozen banks + learned aggregators

Audited positive sets P
All matching category instances
IoU ≥ 0.5; skip uncertain / no-positive

Vo, Vr: B × N × 512

Existing 3-layer parallel decoder
Final-layer H: B × L × 512
Object: Q=H, K/V=Vo; relation: Q=H, K/V=Vr
Same vector gates + FFN

Prefix IDs
Embedding + positions
TRAIN: reference shift-right
Inference: BOS / generated prefix

same attention used by forward

Existing vocabulary head
B × L × 10,000
Next-token distribution

ADDED training-only loss
Actual pre-dropout object attention
A: B × 8 × L × N → head mean a
Lground = −log Σ(n in P) a[n]

Existing LXE
Next-token targets

Inference: beam 5
Generated caption

L = LXE + λ Lground
λ = 0.1 proposed for pilot
No new inference module